MATH& 142 — Comprehensive Final Examination

Original assessment, Modules 1–10

Name: _____ Date: _____

Instructions: Show work and justify results. Supply exact answers unless a decimal approximation is requested. Use radians unless a degree symbol appears. Ten questions, 10 points each, total 100 points.

1. **Radian measure and the unit circle.** Convert 240° to radians. Find $\sin(4\pi/3)$, $\cos(4\pi/3)$, and $\tan(4\pi/3)$ exactly.

2. **Graph transformations.** For $y = -2\sin(3(x - \pi/6)) + 1$, determine amplitude, period, phase shift, midline, and range.

3. **Inverse functions and identities.** Evaluate $\arcsin(\sin(5\pi/6))$. Prove $(1-\sin^2 x)/\cos x = \cos x$ on its domain.

4. **Trigonometric equations.** Solve $\cos(2x) = 1/2$ for all x in $[0, 2\pi)$.

5. **An oblique triangle.** Two sides of a triangle have lengths 6 and 10, with included angle 120°. Calculate the length of the opposite side exactly.

6. **Vectors and dot products.** For $\mathbf{u} = \langle 3, 4\rangle$ and $\mathbf{v} = \langle 4, -3\rangle$, determine $\mathbf{u}+\mathbf{v}$, both magnitudes, and the angle between the vectors.

7. **Complex numbers and rotation.** Let $z = 2(\cos 30^\circ + i \sin 30^\circ)$ and $w = 3(\cos 60^\circ + i \sin 60^\circ)$. Find the product in polar and rectangular forms.

8. **Parametric curves and conics.** Eliminate t from $x = 4\cos t$, $y = 3\sin t$. Identify the resulting conic, its semiaxes, and its foci.

9. **Sequences and series.** A geometric sequence has $a_1 = 9$ and ratio $q = 1/3$. Find a_6 and the infinite sum $\sum_{n=1}^{\infty} a_n$.

10. **Course synthesis.** A point starts at $z_0 = 4$ and obeys $z_{n+1} = \frac{1}{2} i z_n$. Determine z_1, z_2, z_3, then give a formula for z_n. Explain the geometric pattern, including the behavior of magnitude and direction.